

# Who Builds in America: Homebuilders and Housing Supply

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- Housing affordability has become a major concern in the United States, as home prices have generally risen more than household incomes.
- Real prices grew similarly to the 2000's housing boom, but new production was just over half of the 2000's boom.
- There seems to be a bottleneck in the housing supply.
- **Research Question:** What is the business of residential construction and how does it change before and after the bust?

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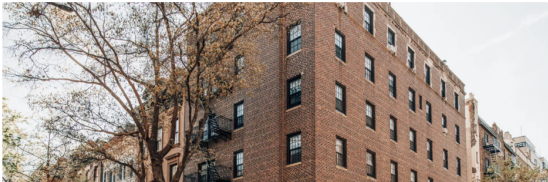


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NEWS

## In 233 U.S. Cities, Even a Starter Home Costs \$1 Million

Half of all states have at least one city where entry-level homes top \$1 million — a lasting effect of pandemic-era price growth.



- Housing supply is typically studied through the lens of supply elasticities, which relate quantity growth to price changes,
  - These elasticities are estimated using reduced-form and structural methods, and fell after the Great Recession.
  - Many of these elasticities explain little about the past 25 years (Louie et al., 2025)
- Little is known about the mechanics of residential construction in the United States, and understanding the sources of disruption to housing markets has large welfare implications
- **This presentation:** Describe the business of residential construction and present new stylized facts on the business of residential construction in the United States.
  - Large builders build cheaper, more similar homes.
  - Large builders dominate the aggregate production of single-family homes. Small firms contracted during the crash, and broadly did not recover to 2000 levels.
  - Entry has fallen, even with consistent price increase.
  - These patterns are consistent with credit constraints, which matter more to small firms.
- **Future:** Develop a model with developer and geographic heterogeneity.

- **Housing Supply:** Housing supply (Dipasquale, 1999), house price changes (Mayer and Somerville, 2000b), land constraints (Saiz, 2010), regulatory barriers (Mayer and Somerville, 2000a) (Gyourko, Saiz, and Summers, 2008) (Gyourko, Hartley, and Krimmel, 2021), elasticities (Aastveit, Albuquerque, and Anundsen, 2023) (Baum-Snow and Han, 2024), and the closing of the frontier (Glaeser and Gyourko, 2025),
  - **Contribution:** New facts about how residential construction firms operate
- **What's Different:** Competition (Quintero, 2022), lower elasticities and regulation (Aastveit et al., 2023), labor supply shocks (Le, 2025), and credit frictions (Gabrovski and Ortego-Marti, 2025)
  - **Contribution:** Show shifts in the market structure of residential construction.
- **Macro-Housing:** Housing IS the business cycle (Leamer, 2007), loose credit (Justiniano et al., 2019), credit matters for small firms (Chodorow-Reich, 2013) (Ayres and Raveendranathan, 2023) (Gabrovski and Ortego-Marti, 2025), large firms (van Straelen, 2023)
  - **Contribution:** Provide evidence consistent with credit constraints mattering for small firms.



# Speculative builders' activities vary by size

- Most new single-family homes in the US are built by “speculative builders.” Speculative builders build homes before having found a buyer.
- Builders need land, labor, and financing to build homes.
- Builders operations differ by size.

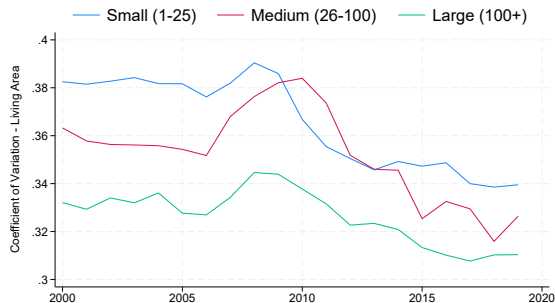
## Small firms:

- Build 1-25 homes in a year
- 97% of active firms in 2019
- Build a variety of houses on smaller lots
- Land is bought before the construction loan is acquired
- Finance with short-term construction loans or cash
- Typically do not have a mortgage arm, and prefer cash and conventional loans as payment

## Large firms:

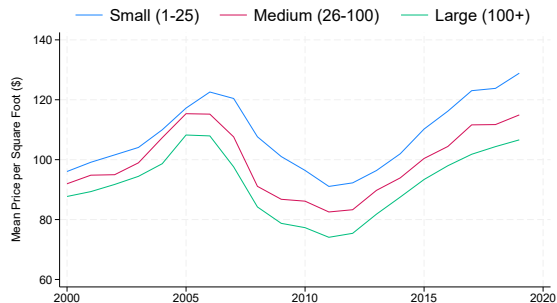
- Build more than 100 homes in a year
- <1% of active firms in 2019
- Build homes from standard plans on large tracts
- Land is acquired either through purchases or land options
- Finance with lines of credit, homebuilder notes, or cash
- Typically can issue mortgages through a financing arm, and take mortgages as payment

# Large builders build similar, cheaper homes



**Figure:** Coefficient of Variation of Living Area by Firm Size

Notes: This plot shows the coefficient of variation by firm size category over time.



**Figure:** Mean Price per Square Foot by Firm Size

Notes: This plot shows the mean real price per square foot by firm size category over time. Adjusted to real terms using CPI excluding shelter.

## Summary Statistics

|                   | Mean     | 50th percentile | Standard deviation | Minimum value | Maximum value |
|-------------------|----------|-----------------|--------------------|---------------|---------------|
| living_area       | 2405.636 | 2255            | 833.1537           | 990           | 6187          |
| number_bedrooms   | 3.578267 | 4               | .8008963           | 1             | 6             |
| number_bathrooms  | 2.911179 | 3               | .8048921           | 1             | 6             |
| sale_amount_real  | 239346.1 | 206670.1        | 127461.2           | 31569.56      | 1233250       |
| ppsf              | 100.5856 | 90.89122        | 42.58002           | 7.807833      | 908.7827      |
| price_per_bedroom | 68099.6  | 58764.17        | 36349.64           | 6848.796      | 1152293       |

Table: Building Facts Summary Statistics

Notes: This table shows descriptive statistics for the variables examined in this section. These data use the Cotality new construction data as the base, but keep observations that are not missing living area, the number of bedrooms, the number of bathrooms, and the price per square foot.

# Large builders dominate the aggregate, but are relatively dispersed

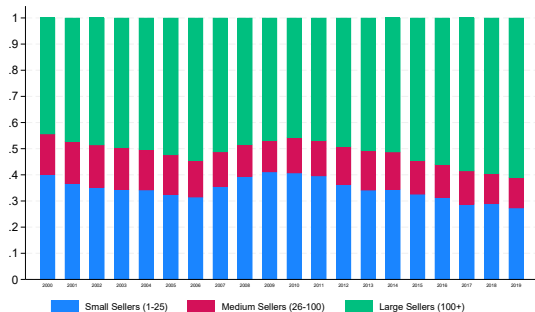


Figure: Aggregate Market Shares by Size

Notes: This plot shows the number of homes produced by each firm-size segment in each year using Cotality data.

►► Size Count Plot

►► Large Builder Share and Change (2000)

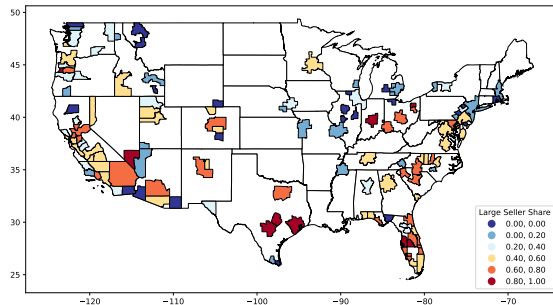


Figure: Large Builder Market Share (2019)

Notes: This map shows the share of homes sold by the large sellers for observed CBSAs in 2019 using Cotality data.

## In aggregate, large builders recover and small builders do not



Figure: Relative Production by Firm Size

Notes: This plot shows the relative number of homes (2000=1) produced by each firm-size segment in each year using Cotality data.

# Large sellers mostly recover but small sellers do not

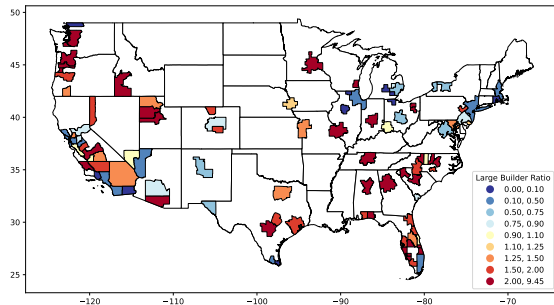


Figure: Relative Production for Large Builders

Notes: This plot shows the production by large firms in 2019 relative to 2000 using Cotality data.

►► Relative Production Distribution

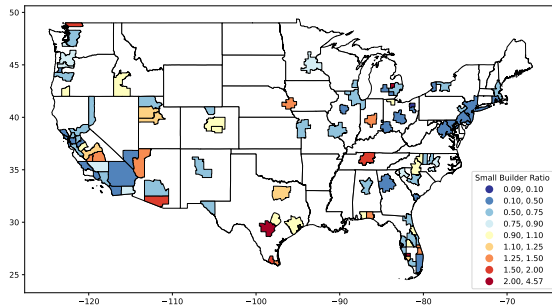
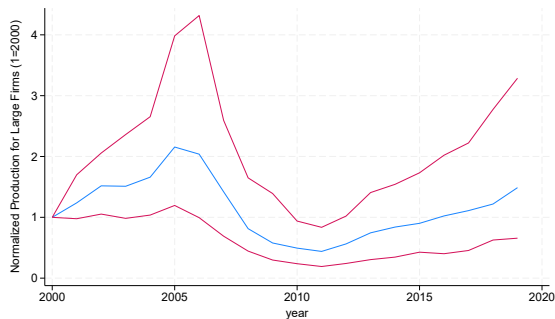


Figure: Relative Production for Small Builders

Notes: This plot shows the production by small firms in 2019 relative to 2000 using Cotality data.

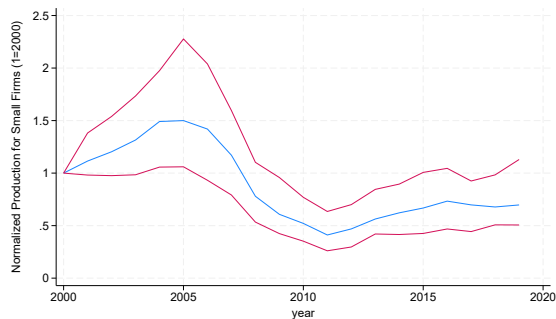
# Large sellers mostly recover but small sellers do not



**Figure:** Relative Production for Large Builders

Notes: This plot shows the production by large firms in 2019 relative to 2000 using Cotality data.

[▶▶ Back to 2019 Maps](#)



**Figure:** Relative Production for Small Builders

Notes: This plot shows the production by small firms in 2019 relative to 2000 using Cotality data.

# Case Study: Akron vs Providence

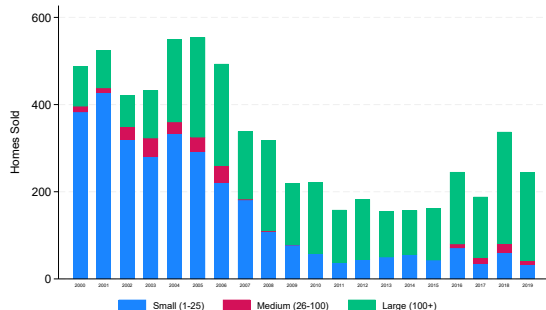


Figure: Akron Production

Notes: This plot shows the number of homes produced by each firm-size segment in Akron, OH by year using Cotality data.

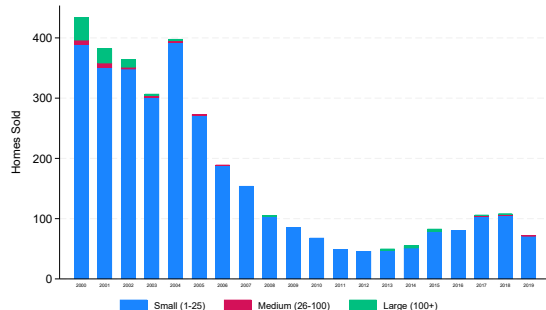


Figure: Providence Production

Notes: This plot shows the number of homes produced by each firm-size segment in Providence, RI by year using Cotality data.



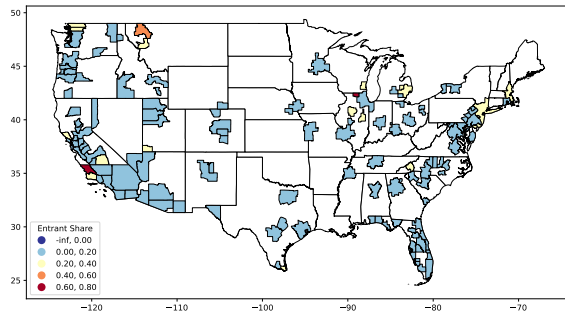


Figure: Caption

# Entry has Fallen

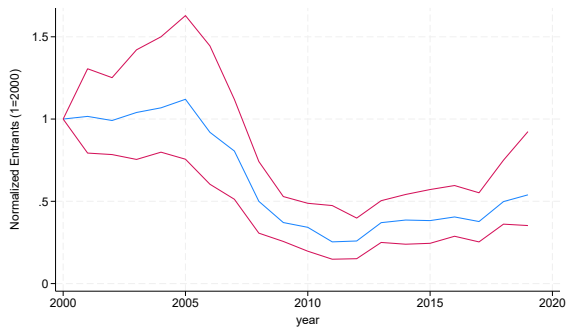


Figure: Caption

# Entry has Fallen



Figure: Caption

# Entry has Fallen



Figure: Caption

Preliminary results (2–5 slides): Present any main findings or progress, such as a central theorem, key graph, or significant regression result, that you have achieved thus far.

- Two builder types, one with a credit constraint and the other who needs to individually search for funding.
- Large builders build one type of home and home quality, and small builders build a range of sizes and qualities.
- High- and low-income consumers purchase homes

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# Market Shares by Size

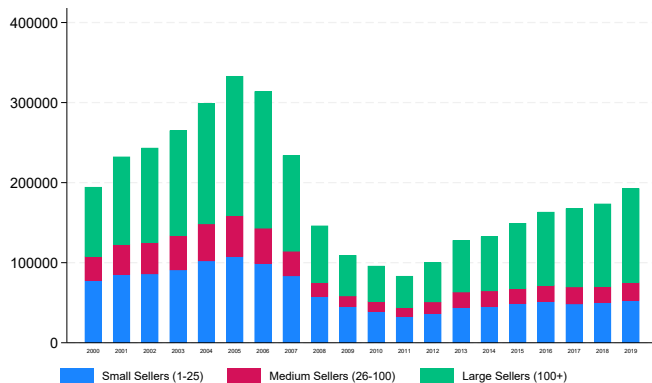


Figure: Home Production by Firm Size

Notes: This plot shows the number of homes produced by each firm-size segment in each year using Cotality data.

# Market Shares by Size

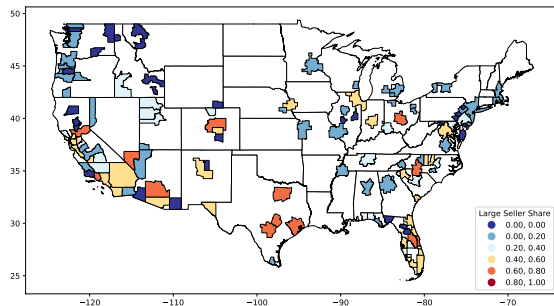


Figure: Large Builder Market Share (2000)

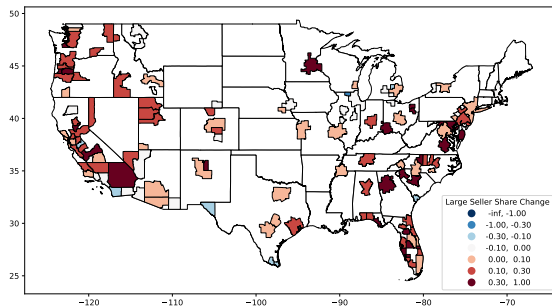


Figure: Large Builder Market Share Change (2000-2019)

► Back to Large Sellers